Part-A

Q1

Ans 3. (2)

Q2

Ans Femur. (2)

Q3

Ans closing of suture. (2)

Q4

Ans Male Skull. (2)

Q5

Ans Identifying individual. (2)

Q6

Ans Head shape. (2)

Q7

Ans Facial recognition. (0)

Q8

Ans All of the above. (2)

Q9

Ans Age at death. (2)

Q10

Ans They are unique to each individual. (2)

Part B

Q 11.2 (4)

Ans Significance of Anthropology in forensic Science.

(1) Anthropology help to determine the age of a person during death with the help of bones.

(2) Forensic Anthropology also help to determine the sex of a person with the help of bone remains.

(3) Forensic anthropology also help in forensic in way like personal identifaction with the help of past physical trauma & injuries.

(4) Anthropology is also help to identify the race of a person with the help of their remains of body (Bones).

(5) Forensic anthropaly help in the cases of mass desasters because with the help of anthropology we identify the person identity which is used in this desasters cases.

(6) Forensic anthropology help in the cases of person identifjaction in the cases of missing person because with the help of it we can identify proble with their special marks (Birth marks, special mark, occupational marks etc)

(7) Forensic anthropology also help in the past ancesto -rs identification or it help to identify the ansistory of a person with their personal traits.

(8) Forensic anthropology help in research purpose because with the help of it we identify person traits & ansistory with their belongings.

Q 12.1 (4.5)

Ans * Chatacterstics features of Male skull *

(1). General apparence of the male skull is more robust & the size of male skull is big as compare to the female.

(2). The cranial capacity of the male skull is more than the female skull.

(3). The orbits of male skull is rounded ~~with~~ ~~smoot~~ margins & the shape of orbits is ~~Square~~

(4). Male skull glabella is more prominent then the female.

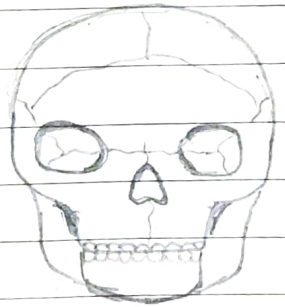
(5). Male skull zygomatic bone is more prominent & the zygomatic arch also prominent in male as compare to female.

(6) The shape of the chin in male is squre shape, so the overall shape of mandble of a male is U - shaped.

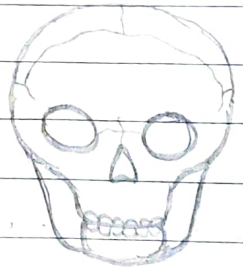
(7) The temporal bone of male is more prominent as compared with females.

(8) The supra orbital magnum & margines is more prominents in male as compared to the females.

(9) The forehead of male is move vertical as compared to the female because female having rounded forehead.



Male Skull



Female Skull.

Q13.2 (4.5)

Ans Somatometry → Soma means body metry means measurements so it means that the measurement of body is know as Somatometry & the word Somatometry comes from Greek word.

* Process of Somatometry →
  - First we collect the all given sample in a very approperate manner which doesn't affect their integrity.
  - carefully measure the sample and write it into their observation notes.
  - care handle the sample & care use instrument on the given sample.
  - After examine the sample properly preserve it for further examination & analysis.

* Forensic significance of measuring parameter in Somatometry *

(1) In Somatometry measuring parameters like head, face,

nose etc, helps to identify the person hight & also help to identify the weight & Apperance of a person

(2) Somatometry also help to reconstruct of facial feature which help to identify the apperance of a person face & help to recreatin the face of a person.

(3) In medico legal cases some times small fragment -ed bone are found in that case with the help of bone peaces we estimate the person heigh & sex of a person.

(4) Somatometry help to identify the race of a person withe the help of deferent measurment like nose, ears, head.

(5) Somatometry help the medico legal cases in various ways like to identify their ancistar traits, or hight or facial feature race, the occupation & other feature which is help the investigati -on in various ways.

Q14.1 (4)

Ans

Portrait Parle/Bertillon Sys.	Modern biometric method
(1) In this we identify & write all physicaly appear features like, shape, size eye, nose ear..	(1) In this we got all information through biometric of a person like shape, size, colour eye, nose & other extra features.
(2) It is time taking process because we identify all feature by phylogical.	(2) In this no time taking process because it gives all data sudden.
(3) In this we wrote only physical appeared features	(3) But in this we got all features which is add in this.
(4) Portrait parle/Bertillon system is very old & tradition method which contain some certain feature	(4) But this is modern technique which is accurate & time benificial.

* Strength & Limitation *

1). Modern biometric methods contain all relivent data which is benefical for forensic investigation. But on the other hand portrait parle having some limitation related to biometrics.

(2) Modern methods data is encryped form which is benifical for transfuring with labs but portrait parle is a writine or physical task.

(3) Modern method is not a time taking process but portrait parle is time taking but it clearly mention all features.

(4) Modern method is very costly technique but portrait parle is budget trendy and easy to under -tand.

Q 15.2 (5)

Answer Facial reconstruction → In this we reconstruct the skull & identify the facial features & apperance of a person.

(1) facial identification
(2) facial restoration
(3) facial reconstruction.

Facial reconstruction technique

* Superimposition technique
* 2D reconstruction
* 3D reconstruction

* Super imposition technique → In this we having antimort-em picture & post mortem picture of a person & superimpose it to identify person facial feature & apparence.

(1) Fading or Bluring → In this we having 2 camera placed on the skull & picture of a person. then superimpose it & with the help of fading try to identify features.

(2) Swaping technique → In which we placed photo of skull onto the person picture like digonaly, vertically, rectanga

box, & Horizontal & match the features.

* 2D Technique → In this we draw or 2-D reconstruct the facial feature in this two methode is used photographic or. videographic, X-ray also inclu-ded

(1) Photographic tech. → In this with the help of camera we do superimposition of the person picture

(2) Video graphic tech. → In the superimposition tech. clearly mention this tech.

(3) X-ray → with the help of it we can identify the person facial feature with the skull.

* 3D technique → In this we do 3-Dimensional inspecti-on & reconstruct the face.
In this we made sculpture of a person

(1) Muscles to muscles Method (Europion method) we connect muscles to the skull and construct face.

(2) Tissue depth method (American) method → In this with the help of tissue we do the reconstruction of face.

(3) Combined method → This technique is more appropreate result-

Part-C

8

Q 16.2

Ans

Human skeleton System

- Axial System (80)
  - Skull (29)
  - vertibral (26)
  - Thrasic cage (24)
  - Strenum (1)
- (126) Appendicular System
  - Limbs (120)
  - Gridles (6)

Appendicular syster

- Limbs (120)
  - upper limbs (60)
  - lower limbs (60)
- (Gridles) 6
  - pactorial gridles (4)
  - pelvic gridles (2)

Types of bone →

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(1) flat bone
(2) Sysmoid bone
(3) short bone
(4) long bone
(5) Irregular bone
(6) Numotic bone

(1) Flat bone → which is flat surface bone in body like Stremum, illum

(2) Short bone → which is having same length & breadth short in nature like tarcles & carples.

(3) Neumotic bone → In this all skull bone is Neumotic bone (Skull bones)

(4) Long bone → In this length is more breadth is less & long in nature like Humerus, Femur.

(5) Irregular bone → which is irregular is shape like Vertebral.

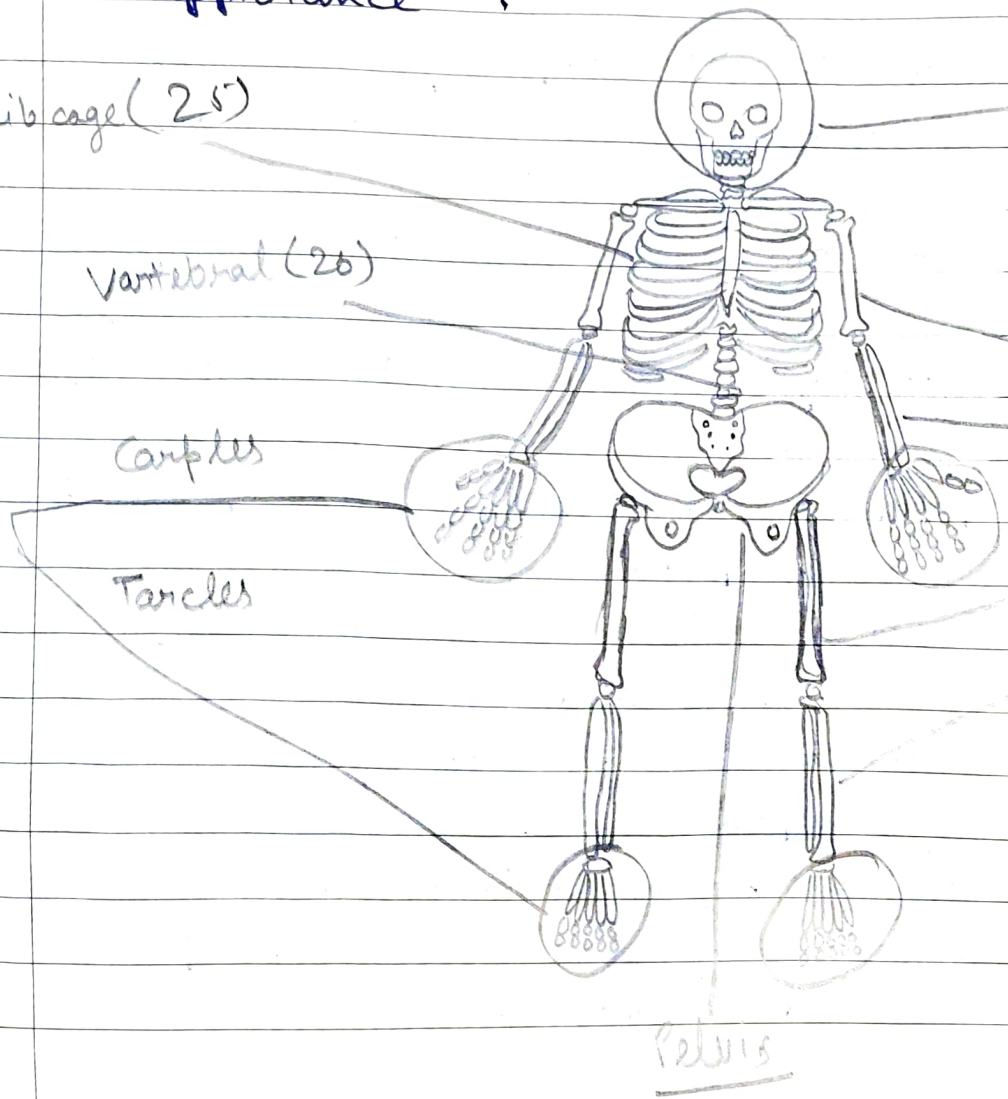
(6) Sysmoid bone → In the flared bone is present like skull bones etc.

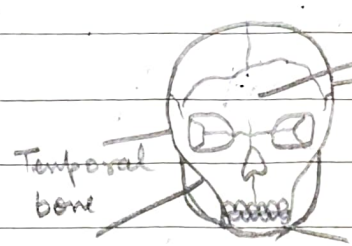
* Two type of bones *

(1) Compact bone (2) Spongy bone.

(1) Compact bone → are generally larger in size & having rough surface like femur, skull.

(2) Spongy bone → are generally smaller in size & smooth apperance.





Skull
29 bones

Upper Limbs

Limbs 120

Lower Limbs

Q17]

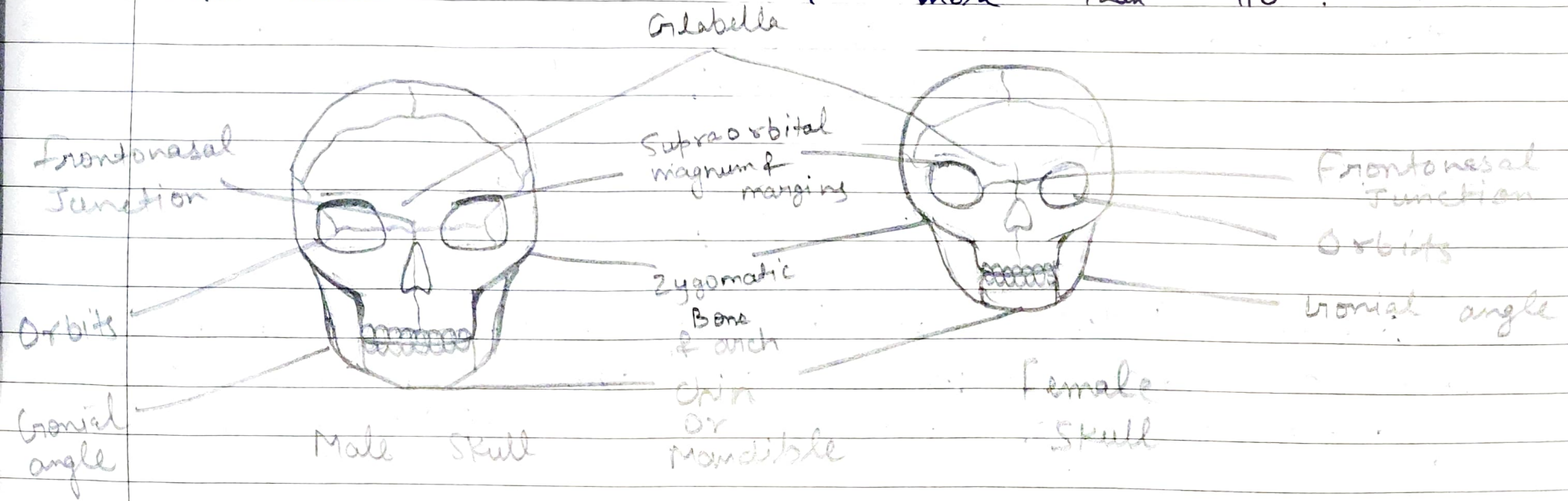
(8)

Sex from Skull

Ans

Male	Female
(1) General apparence is rough.	(1) General apparence is smooth
(2) Cranial capacity is more	(2) Cranial capacity is less
(3) Orbits is square with round margins	(3) Orbits is oval & sharp margins
(4) Zygomatic bone is more prominent	(4) Zygomatic bone is less prominent
(5) Zygomatic arch is more defined	(5) Zygomatic arch is less defined
(6) palate shape is U-Shaped	(6) Palet shape is parabolic shaped.
(7) Mandible is generaly U-shaped	(7) Mandible is generally V-shaped
(8) Mendublar condyle is more prominent	(8) Mendibular condyle is less prominent.
(9) Supraorbital magnum & margines are prominent in male	(9) supra orbital magnum & margines are less prominent in females.

(10) Temporal bone are more prominent.	(10) Temporal bone are less prominent
(11) Forehead are more vertical	(11) forehead of a female is rounded
(12) Glabella of a male is more prominent	(12) Glabella of a female is less prominent
(13) Frontonasal junction is more curved	(13) Frontonasal junction is more pronounced in female.
(14) Gonial angle is 90-110° in male	(14) Gonial angle in female is more than 110°.



Q18.2

Answer * Cephalic index → It is defined as head breadth from both the parital bone is divided by head length from glebla to opisthocranion, multiplied by 100

Formula ⇒ $\frac{\text{head breadth}}{\text{head length}} \times 100$

* nasal index → It is defined as nose breadth from alare to alare is divided by nose length nason to subnasal & multiplied by 100.

formula ⇒ $\frac{\text{nose breadth}}{\text{nose length}} \times 100$

* Cranial index → It is defined as head breadth from the both temporal is devided by head length glabella to Opisthocranion. is multiplied by 100

formula → $\frac{\text{head breadth}}{\text{head length}} \times 100$

* Upper facial indices → In this the length of the mid incisors from mandible to nason is devided by breadth of the face from zygomatic arch distances of both is multiplied by 100

formula → $\frac{\text{face length}}{\text{face breadth}} \times 100$

* Anatomical variation to ethnic or regional backgrounds *

(1) With the help of all these index we identify the overall facial measurment of a person & identify the race, & also match with this it to their belongings.

(2) All these index help us to identify the past varation of the body because of there habit & there living style so with all these this measurement help to identify the changes due to climate & lifestyle.

* Nasal idex is help to identify the measurment of the nose size & shape with the help of these we also identify the varation of there nose shape due to climate & life style.

* cranial index is help to identify the overall shape & size of the head with the help of these we identify the race of a person & there ancistory.

* facial indices is also define the face shape & the size of the face with the help of these we identify how the varation occure & which race belonged person is this.

Q19.1 (15)

Ans facial Super imposition techniques → There are so many facial super imposition techniques to examine & identifying the person identity.

Techniques → (1) 2D superimposition technique
(2) 3D superimposition technique

(1) 2-D superimposition technique → In this there are three methode
(1) Photographic technique
(2) Videographic technique
(3) ~~3D scan~~ X-ray scanning

* Photographic → In this we having pictures of antimortem & postmortem them we do super imposition through Vantide, horizontal, digonal, & fading.

* Videographic → In this we having two camera one is place

On the picture & one is placed on the skull then with the display we fade or blurring the picture to identify the facial features.

(*) X-Ray → with the help of this we can identify the X-Ray image of skull & muscles attachment

(2) 3-D superimposition technique → In this we made a sculpture of the man to carefully & deeply examination of facial features. In this we made model with the help of clay.

In this 3 method is used

* Tissue depth method → In this method as known as American method in this we made a model with tissue to tissue connection.
* Muscles to Muscles → In this method also know as europian method in this muscles to muscles connection technique is used.
* Combined method → In this we used both method for better result & accuracy.

* Strength & Limitation →

(1) 2-D superimposition technique is best for visual examination but it is not give us appropriate result.

(2) 3D superimposition technique give us a very refined facial reconstruction which is very benificial but this method is time take

* Uses *

(1) The superimposition technique is used in both confirmed & disputed cases because it give us proper result

(2) With the help of these techniques we identify the person from the skull in deep & give opinion on the basis of result which help to solve criminal cases.

(3) In disputed identity cases person some time having special feature with the help of superimposition we clearly identify it side by side

Q 20.1

7.5

Ans * Role of forensic Odontology in Mass disaster *

(1) Some time in Mass disaster cases we did not identify the person body but forensic Odonto -logis easily identify it with the help of teeth.

(2) Some mass disaster cases is very difficult to identify the age of a person but with the help of odontology & eruption of teeth help to identify age.

(3) Occupation mark are present on the teeth when a mass disaster cases after with the help of person ocupation we easily identify person body.

(4) Some time in mass disaster case we having tooth sample so we can also extract DNA from it & identify person identity & Match with it to the victim.

(5) ~~when~~ some time tear & grooves marks help to identify the body a person.

& there chewing pattern & also identify the person locality & religion.

* Dental record.

(1) With the help of dental record we are easily identify the person locality & habits & also identify the person occupation.

(2) Some time with the help of dental record we can identify the Dental anomiles & any special feature which is present in human teeth so we can easy match it to the victim

(3) Sometime dental recode & dental teeth having contain some of the part of DNA which help to connect it to the victim directly & also identify the race of chewing habits of a person

(4) Dental records are very crucial because it having all the details related to the person identity.

End of Writing